آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 10

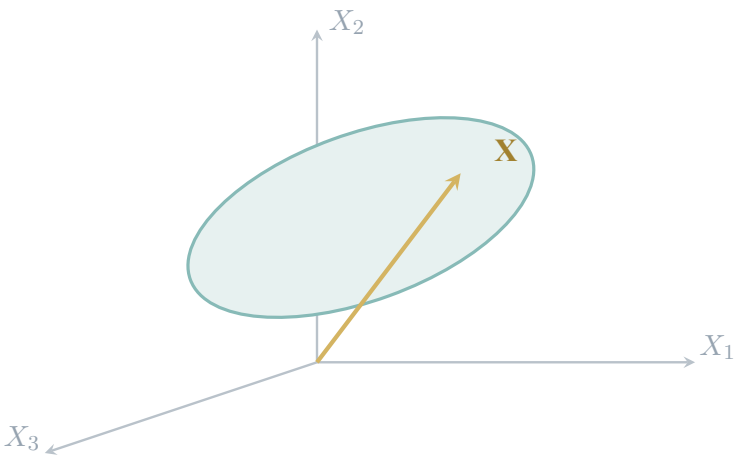
Multiple Random Variables

چند متغیر تصادفی

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Learning Objectives

After completing this module, students will be able to:

1. Extend joint distributions to N random variables
2. Define and work with random vectors
3. Compute mean vectors and covariance matrices
4. Apply the multivariate Gaussian distribution
5. Perform linear transformations of random vectors

10.1 Introduction: From 2 to N Variables

Real systems often involve many uncertain quantities simultaneously:

- **Sensor array:** N sensors each producing a noisy measurement
- **MIMO system:** $M \times N$ channel coefficients
- **Parameter estimation:** p unknown parameters
- **Signal samples:** N consecutive noise samples

We need a framework for **N-dimensional** random vectors.

10.2 Joint Distributions for N Random Variables

Joint PDF

For $X_1, X_2, \ldots, X_n$:

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) \geq 0, \quad \int \cdots \int f_{X_1,\ldots,X_n}\, dx_1 \cdots dx_n = 1$$

Marginal Distributions

Integrate (or sum) out unwanted variables:

$$f_{X_1}(x_1) = \int \cdots \int f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)\, dx_2 \cdots dx_n$$

Conditional Distributions

$$f_{X_1|X_2,\ldots,X_n}(x_1|x_2,\ldots,x_n) = \frac{f_{X_1,\ldots,X_n}(x_1,\ldots,x_n)}{f_{X_2,\ldots,X_n}(x_2,\ldots,x_n)}$$

Mutual Independence

$X_1, \ldots, X_n$ are mutually independent if:

$$f_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \prod_{i=1}^{n} f_{X_i}(x_i)$$

10.3 Random Vectors

Definition

A **random vector** is an ordered collection of random variables:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

Engineering Examples

Random Vector	Components	Context
Sensor array output	$X_1, \ldots, X_n$ = sensor readings	Array processing
MIMO channel	$h_1, \ldots, h_n$ = channel gains	Wireless communications
Noise vector	$N_1, \ldots, N_n$ = noise samples	Signal processing
State vector	Position, velocity, acceleration	Kalman filtering
Feature vector	Extracted measurements	Pattern recognition

10.4 Mean Vector

Definition

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_n] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

Properties

- $E[AX + b] = \mathrm{AE[X]} + b = A\mu + b$
- Linearity extends to vectors and matrices

10.5 Covariance Matrix

Definition

$$\boldsymbol{\Sigma} = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

Element (i, j): $\Sigma_{ij} = \text{Cov}(X_i, X_j)$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \sigma_2^2 & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \cdots & \cdots & \sigma_n^2 \end{bmatrix}$$

Properties

1. **Symmetric:** $\Sigma = \Sigma^T$
2. **Positive semi-definite:** $v^T \Sigma v \geq 0$ for all v
3. **Diagonal = variances:** $\Sigma_{ii} = \text{Var}(X_i)$
4. **Off-diagonal = covariances:** $\Sigma_{ij} = \text{Cov}(X_i, X_j)$
5. **Independent components $\rightarrow$ diagonal** Σ

Alternative Formula

$$\boldsymbol{\Sigma} = E[\mathbf{X}\mathbf{X}^T] - \boldsymbol{\mu}\boldsymbol{\mu}^T$$

10.6 Multivariate Gaussian Distribution

Definition

$X \sim N(\mu, \Sigma)$ has PDF:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

where $|\Sigma|$ = determinant of Σ.

Key Properties

1. **Marginals are Gaussian:** Any subset of components is jointly Gaussian
2. **Conditionals are Gaussian:** $X_1 \mid X_2 = x_2$ is Gaussian
3. **Linear transformations:** $Y = \mathrm{AX} + b \sim N(A\mu + b,\ A\Sigma A^T)$
4. **Uncorrelated = Independent** (unique to Gaussian!)
5. **Fully specified** by μ and Σ (only 2 parameters needed per variable plus correlations)

Why So Important?

- Central Limit Theorem: sums of many RVs $\rightarrow$ multivariate Gaussian
- Thermal noise vectors are multivariate Gaussian
- Analytically tractable (closed-form marginals, conditionals)
- Optimal filters (Kalman, Wiener) assume Gaussian

$2D$ Case (Bivariate Gaussian)

For $n = 2$ with $\mu = [0,\ 0]^T$, $\sigma_1 = \sigma_2 = 1$:

$$f(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right)$$

Contours are **ellipses** (tilted when $\rho \neq 0$).

10.7 Linear Transformations of Random Vectors

Transformation $Y = AX + b$

If X is a random vector with mean μ_X and covariance Σ_X:

$$E[\mathbf{Y}] = A\boldsymbol{\mu}_X + \mathbf{b}$$

$$\boldsymbol{\Sigma}_Y = A\boldsymbol{\Sigma}_X A^T$$

Engineering Applications

Beamforming: Output $y = w^T X$ (weight vector applied to sensor array)

- $E[y] = w^T \mu$
- $\text{Var}(y) = w^T \Sigma w$

Whitening: Find matrix W such that $W\Sigma W^T = \text{I}$ (decorrelates data)

- $W = \Sigma^{-1/2}$

Principal Component Analysis: Rotate to diagonalize Σ

10.8 Practice Problems

Problem 1

Random vector $X = [X_1,\ X_2,\ X_3]^T$ with $\mu = [1,\ 0,\ -1]^T$, and covariance matrix $\Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$.

(a) Find $\text{Var}(X_1 + X_2 + X_3)$. (b) Find $\text{Cov}(X_1,\ X_2 + X_3)$. (c) Are X_1 and X_3 uncorrelated?

Solution

(a) $\text{Var}(X_1 + X_2 + X_3) = 1^T \Sigma 1 = 4 + 2 + 3 + 2(1) + 2(0) + 2(-1) = 9 + 0 = 9$
Wait: $= \sum_{ij}$ all entries $= 4 + 1 + 0 + 1 + 2 + (-1) + 0 + (-1) + 3 = 9$

(b) $\text{Cov}(X_1,\ X_2 + X_3) = \text{Cov}(X_1,\ X_2) + \text{Cov}(X_1,\ X_3) = 1 + 0 = 1$

(c) $\text{Cov}(X_1,\ X_3) = 0$, so yes, X_1 and X_3 are uncorrelated.

Problem 2

$X \sim N(0,\ \Sigma)$ with $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$. Find the conditional distribution of $X_1 \mid X_2 = x_2$.

Solution For bivariate Gaussian:

$$X_1 \mid X_2 = x_2 \sim N\left(\mu_1 + \rho\left(\frac{\sigma_1}{\sigma_2}\right)(x_2 - \mu_2),\ \sigma_1^2(1 - \rho^2)\right)$$

$$= N(\rho x_2,\ 1 - \rho^2)$$

The conditional mean is a linear function of x_2, and conditional variance is reduced by factor $(1 - \rho^2)$.

Next Module: Module 11 — Central Limit Theorem